Complex Analysis Homework 8

1. Find the Laurent series of $\frac{1}{z+1}+\frac{1}{z+i}$ at 1 from $(z-1)^{-2}$ to $(z-1)^{2}$ in each of the following domains.

(a) the region $0<|z-1|<\sqrt{2}$;

(b) the region $\sqrt{2}<|z-1|<2$;

(c) the region $|z-1|>2$.

Solution

$$f(z)=\frac{1}{z+1}+\frac{1}{z+i}=\frac{1}{(z-1)+2}+\frac{1}{(z-1)+1+i}$$

(a) $0<|z-1|<\sqrt{2}$;

$$f(z)=\frac{1}{2\left(1+\frac{z-1}{2}\right)}+\frac{1}{(1+i)(1+\frac{z-1}{1+i})}$$

$$=\frac{1}{2}\left(1-\frac{z-1}{2}+\frac{(z-1)^2}{2^2}-\frac{(z-1)^3}{2^3}+\cdots\right)+\frac{1}{(1+i)}\left(1-\frac{z-1}{1+i}+\frac{(z-1)^2}{(1+i)^2}-\frac{(z-1)^3}{(1+i)^3}+\cdots\right)$$

$$=\left(\frac{1}{2}+\frac{1}{(1+i)}\right)-\left(\frac{1}{2^2}+\frac{1}{(1+i)^2}\right)(z-1)+\left(\frac{1}{2^3}+\frac{1}{(1+i)^3}\right)(z-1)^2-\cdots$$

(b) $\sqrt{2}<|z-1|<2$;

$$f(z)=\frac{1}{2\left(1+\frac{z-1}{2}\right)}+\frac{1}{(z-1)(1+\frac{1+i}{z-1})}$$

$$=\frac{1}{2}\left(1-\frac{z-1}{2}+\frac{(z-1)^2}{2^2}-\frac{(z-1)^3}{2^3}+\cdots\right)+\frac{1}{(z-1)}\left(1-\frac{1+i}{z-1}+\frac{(1+i)^2}{(z-1)^2}-\frac{(1+i)^3}{(z-1)^3}+\cdots\right)$$

$$=\cdots-\frac{(1+i)}{(z-1)^2}+\frac{1}{(z-1)}+\frac{1}{2}-\frac{z-1}{2^2}+\frac{(z-1)^2}{2^3}+\cdots$$

(c) $|z-1|>2$.

$$f(z)=\frac{1}{(z-1)\left(1+\frac{2}{z-1}\right)}+\frac{1}{(z-1)(1+\frac{1+i}{z-1})}$$

$$=\frac{1}{(z-1)}\left(1-\frac{2}{z-1}+\frac{2^2}{(z-1)^2}-\frac{2^3}{(z-1)^3}+\cdots\right)+\frac{1}{(z-1)}\left(1-\frac{1+i}{z-1}+\frac{(1+i)^2}{(z-1)^2}-\frac{(1+i)^3}{(z-1)^3}+\cdots\right)$$

$$=\cdots-\frac{2+(1+i)}{(z-1)^2}+\frac{2}{(z-1)}$$

2. Evaluate the following integrals with counterclockwise contour using residue theorem.

(a) $\int_{|z|=1}\frac{\sin z}{z^2}dz$

(b) $\int_{|z|=2}\frac{e^z}{z^2-1}dz$

(c) $\int_{|z|=2}\frac{z}{\cos z}dz$

(d) $\int_{|z|=1} \frac{z^4}{\sin z} dz$

(e) $\int_{|z-1|=1} \frac{1}{z^8 - 1} dz$

(f) $\int_{|z-1/2|=3/2} \frac{\tan z}{z} dz$

Solution

(a) Using the residue theorem,

$$\int_{|z|=1} \frac{\sin z}{z^2} dz = 2\pi i \operatorname{Res}\left[\frac{\sin z}{z^2}, 0\right] = 2\pi i \left(\lim_{z\to 0} \frac{\sin z}{z}\right) = 2\pi i\,(1) = 2\pi i.$$

(b) Using the residue theorem,

$$\int_{|z|=2} \frac{e^z}{z^2 - 1} dz =$$

$$= 2\pi i \left(\operatorname{Res}\left[\frac{e^z}{z^2-1}, 1\right] + \operatorname{Res}\left[\frac{e^z}{z^2-1}, -1\right]\right) =$$

$$= 2\pi i \left(\left.\frac{e^z}{2z}\right|_{z=1} + \left.\frac{e^z}{2z}\right|_{z=-1}\right) = 2\pi i \left(\frac{e^1 - e^{-1}}{2}\right) =$$

$$= 2\pi i \sinh 1.$$

(c) Using the residue theorem,

$$\int_{|z|=2} \frac{z}{\cos z} dz = 2\pi i \left(\operatorname{Res}\left[\frac{z}{\cos z}, \frac{\pi}{2}\right] + \operatorname{Res}\left[\frac{z}{\cos z}, -\frac{\pi}{2}\right]\right) =$$

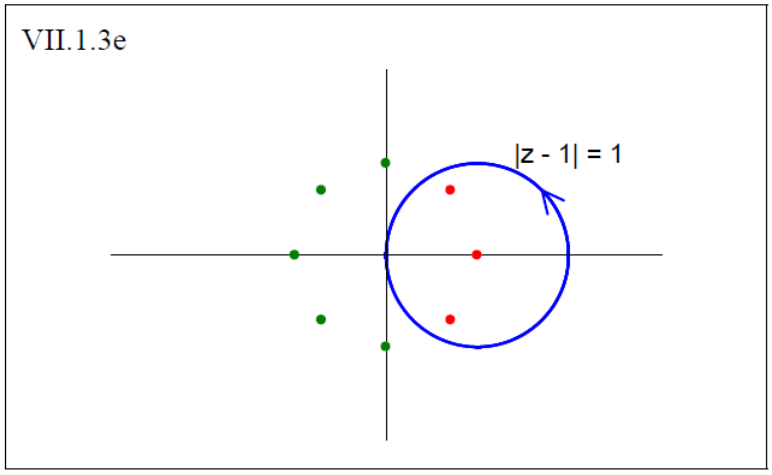
$$= 2\pi i \left(\frac{z}{-\sin z}\bigg|_{z=\pi/2} + \frac{z}{-\sin z}\bigg|_{z=-\pi/2}\right) = 2\pi i \left(-\frac{\pi}{2} - \frac{\pi}{2}\right) =$$

$$= -2\pi^2 i.$$

(d) Using the residue theorem,

$$\int_{|z|=1} \frac{z^4}{\sin z} dz = 2\pi i \operatorname{Res}\left[\frac{z^4}{\sin z}, 0\right] = 2\pi i \left(\frac{z^4}{\cos z}\bigg|_{z=0}\right) = 0.$$

(e)



Using the residue theorem with the contour in Figure VII.1.3e and rule 4,

$$\int_{|z-1|=1} \frac{1}{z^8-1} dz = 2\pi i \left(\operatorname{Res}\left[\frac{1}{z^8-1}, 1\right] + \operatorname{Res}\left[\frac{1}{z^8-1}, e^{i\pi/4}\right] + \operatorname{Res}\left[\frac{1}{z^8-1}, e^{-i\pi/4}\right]\right) =$$

$$= 2\pi i \left(\frac{1}{8z^7}\bigg|_{z=1} + \frac{1}{8z^7}\bigg|_{z=e^{i\pi/4}} + \frac{1}{8z^7}\bigg|_{z=e^{-i\pi/4}}\right) = \frac{\pi i}{4}\left[1 + e^{-\pi i/4} + e^{\pi i/4}\right] =$$

$$= \frac{\pi i}{4}\left[1 + \frac{1-i}{\sqrt{2}} + \frac{1+i}{\sqrt{2}}\right] = \frac{\pi i}{4}\left(1+\sqrt{2}\right).$$

(f) The function $\frac{\tan z}{z} = \frac{\sin z/z}{\cos z}$ has removable singularity at $z = 0$, and a isolated singularity at $z = \frac{\pi}{2}$ inside the domain $|z - 1/2| = 3/2$. Using the residue theorem,

$$\int_{|z-1/2|=3/2} \frac{\tan z}{z} dz =$$

$$= 2\pi i \operatorname{Res}\left[\frac{\sin z/z}{\cos z}, \frac{\pi}{2}\right] = 2\pi i \left(\lim_{z\to\pi/2} \frac{\sin z/z}{-\sin z}\right) = 2\pi i \left(\left.\frac{\sin z/z}{-\sin z}\right|_{z=\pi/2}\right) = (2\pi i)\left(-\frac{2}{\pi}\right) =$$

$$= -4i.$$

3. Using residue theory, calculate the following integrals:

(a) $\int_0^{2\pi} \frac{d\theta}{a + b\sin\theta}, \quad a > b > 0$

(b) $\int_0^{2\pi} \frac{d\theta}{1 + 3\cos^2\theta}$

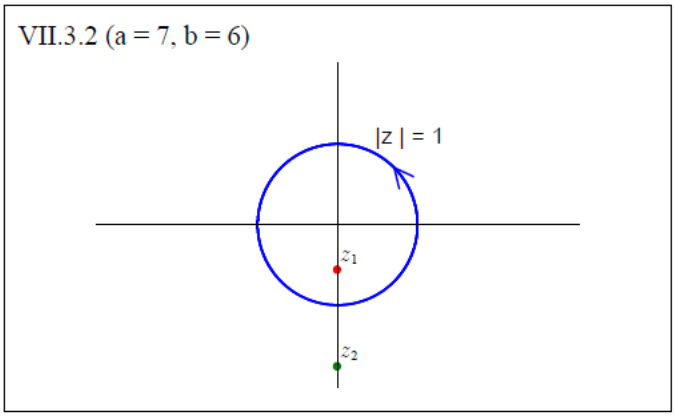
(c) $\int_0^{\pi} \frac{\cos 2\theta}{1 - 2a\cos\theta + a^2} d\theta$ where $a^2 < 1$.

(d) $\int_0^{2\pi} \frac{\cos 2\theta}{5 - 4\cos\theta} d\theta$

(e) $\int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2}$

Solution

(a)



Set

$$I = \int_0^{2\pi} \frac{1}{a + b\sin\theta} d\theta.$$

A change of variables gives

$$I = \int_0^{2\pi} \frac{1}{a + b\sin\theta} d\theta = \oint_{|z|=1} \frac{1}{\left(a + b \cdot \frac{1}{2i}\left(z - \frac{1}{z}\right)\right)} \frac{dz}{iz} = \oint_{|z|=1} \frac{2}{bz^2 + 2azi - b} dz$$

Integrate

$$f(z) = \frac{2}{bz^2 + 2azi - b} = \frac{2}{b\left(z - \left(\frac{-a-\sqrt{a^2-b^2}}{b} i\right)\right)\left(z - \left(\frac{-a+\sqrt{a^2-b^2}}{b} i\right)\right)}$$

along the unit circle contour in Figure VII.3.2.
Residue at a simple pole at $z_1 = \frac{-a+\sqrt{a^2-b^2}}{b} i$,

$$\text{Res}\left[\frac{2}{bz^2 + 2azi - b}, \frac{-a + \sqrt{a^2 - b^2}}{b} i\right] = \left.\frac{2}{2bz + 2ai}\right|_{z=\frac{-a+\sqrt{a^2-b^2}}{b} i} = -\frac{i}{\sqrt{a^2 - b^2}}$$

Using the *Residue Theorem*, we thus obtain

$$I = 2\pi i \cdot \left(-\frac{1}{\sqrt{a^2 - b^2}} i\right),$$

and therefore

$$\int_0^{2\pi} \frac{1}{a + b\sin\theta} d\theta = \frac{2\pi}{\sqrt{a^2 - b^2}}, \qquad a > b > 0.$$

(b) $\displaystyle\int_0^{2\pi}\frac{d\theta}{1+3\cos^2\theta}=\int_{|z|=1}\frac{1}{1+3\left(\frac{z+z^{-1}}{2}\right)^2}\frac{dz}{iz}=\int_{|z|=1}\frac{-4izdz}{3z^4+10z^2+3}$

$z^2=-\frac{1}{3}\quad\text{or}\quad -3$

$$\text{Res}\left(\frac{i}{\sqrt{3}}\right)=\lim_{z\to\frac{i}{\sqrt{3}}}\frac{-4iz}{(3z^4+10z^2+3)'}=\lim_{z\to\frac{i}{\sqrt{3}}}\frac{-4iz}{12z^3+20z}=\lim_{z\to\frac{i}{\sqrt{3}}}\frac{-i}{3z^2+5}=-\frac{i}{4}$$

$$\text{Res}\left(-\frac{i}{\sqrt{3}}\right)=\lim_{z\to-\frac{i}{\sqrt{3}}}\frac{-4iz}{(3z^4+10z^2+3)'}=\lim_{z\to-\frac{i}{\sqrt{3}}}\frac{-4iz}{12z^3+20z}=\lim_{z\to-\frac{i}{\sqrt{3}}}\frac{-i}{3z^2+5}=-\frac{i}{4}$$

$$\int_0^{2\pi}\frac{d\theta}{1+3\cos^2\theta}=2\pi i\left[\text{Res}\left(\frac{i}{\sqrt{3}}\right)+\text{Res}\left(-\frac{i}{\sqrt{3}}\right)\right]=2\pi i\left(-\frac{i}{4}\right)\times 2=\pi$$

c) If $a=0$, then

$$\int_0^{\pi}\frac{\cos 2\theta\,d\theta}{1-2a\cos\theta+a^2}=\int_0^{\pi}\cos 2\theta\,d\theta=\frac{1}{2}\left[\sin 2\theta\right]_0^{\pi}=\frac{1}{2}[0-0]=0.$$

Assume $a\neq 0$. We can write

$$\int_0^{\pi}\frac{\cos 2\theta\,d\theta}{1-2a\cos\theta+a^2}=\frac{1}{2}\int_0^{2\pi}\frac{\cos 2\theta\,d\theta}{1-2a\cos\theta+a^2}=\frac{1}{2}\int_0^{2\pi}\frac{\frac{e^{2\theta i}+e^{2\theta i}}{2}}{1-2a\frac{e^{\theta i}+e^{\theta i}}{2}+a^2}\,d\theta.$$

Let $z=e^{i\theta}$. Then $dz=iz\,d\theta\implies d\theta=\frac{dz}{iz}$. Hence,

$$\begin{aligned}\frac{1}{2}\int_0^{2\pi}\frac{\frac{e^{2\theta i}+e^{2\theta i}}{2}}{1-2a\frac{e^{\theta i}+e^{\theta i}}{2}+a^2}\,d\theta &= \frac{1}{2}\int_{|z|=1}\frac{\frac{z^2+\frac{1}{z^2}}{2}}{1-a\left(z+\frac{1}{z}\right)+a^2}\frac{dz}{iz}=\frac{1}{4i}\int_{|z|=1}\frac{z^4+1}{z^2\left(z-az^2-a+a^2z\right)}\,dz\\ &= -\frac{1}{4ia}\int_{|z|=1}\frac{z^4+1}{z^2\left(z^2-\frac{1+a^2}{a}z+1\right)}\,dz=-\frac{1}{4ia}\int_{|z|=1}\frac{z^4+1}{z^2\left(z-a\right)\left(z-\frac{1}{a}\right)}\,dz\end{aligned}$$

By the Residue Theorem and the fact that $|a|<1$, since $a^2<1$

$$\int_{|z|=1}\frac{z^4+1}{z^2\left(z-a\right)\left(z-\frac{1}{a}\right)}\,dz=2\pi i\left[Res f(f,0)+Res(f,a)\right]$$

where $f(z)=\frac{z^4+1}{z^2(z-a)\left(z-\frac{1}{a}\right)}$. It is easily obtained

$$\begin{aligned}Res(f,0) &= \lim_{z\to 0}\frac{d}{dz}\left[z^2\frac{z^4+1}{z^2\left(z-a\right)\left(z-\frac{1}{a}\right)}\right]=\lim_{z\to 0}\frac{d}{dz}\left[\frac{z^4+1}{\left(z-a\right)\left(z-\frac{1}{a}\right)}\right]\\ &= \lim_{z\to 0}\frac{-4z^3\left(z^2-az-\frac{1}{a}z+1\right)-\left(z^4-1\right)\left(2z-a-\frac{1}{a}\right)}{\left(z^2-az-\frac{1}{a}z+1\right)^2}=a+\frac{1}{a}\end{aligned}$$

and

$$Res(f,a) \;=\; \lim_{z\to a}(z-a)\frac{z^4+1}{z^2\,(z-a)\left(z-\frac{1}{a}\right)} = \lim_{z\to a}\frac{z^4+1}{z^2\left(z-\frac{1}{a}\right)} = \frac{a^4+1}{a^2\left(a-\frac{1}{a}\right)} = \frac{a^4+1}{a(a^2-1)}.$$

So

$$\int_{|z|=1}\frac{z^4+1}{z^2\,(z-a)\left(z-\frac{1}{a}\right)}\,dz = 2\pi i\left(a+\frac{1}{a}+\frac{a^4+1}{a(a^2-1)}\right) = 2\pi i\frac{2a^4}{a(a^2=1)} = -\frac{4\pi i a^3}{1-a^2}.$$

Finally,

$$\int_0^{\pi}\frac{\cos 2\theta\,d\theta}{1-2a\cos\theta+a^2} = -\frac{1}{4ia}\cdot\left(-\frac{4\pi i a^3}{1-a^2}\right) = \pi\frac{a^2}{1-a^2}.$$

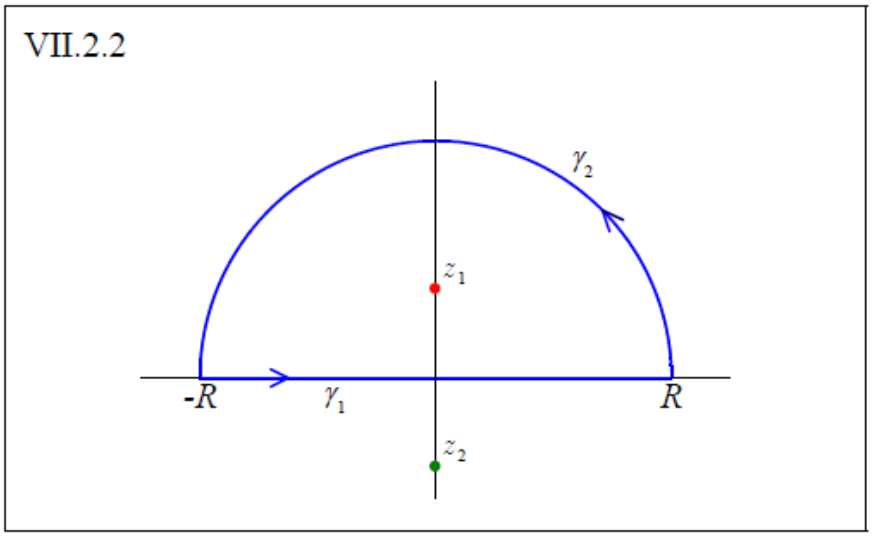
(d) $\displaystyle\int_0^{2\pi}\frac{\cos 2\theta}{5-4\cos\theta}d\theta = \int_{|z|=1}\frac{\frac{1}{2}(z^2+z^{-2})}{5-4\left(\frac{z+z^{-1}}{2}\right)}\frac{dz}{iz} = \int_{|z|=1}\frac{i(z^4+1)dz}{2z^2(z-2)(2z-1)}$

$\displaystyle\text{Res}(0) = \lim_{z\to 0}\frac{d}{dz}\left(\frac{i(z^4+1)}{2(z-2)(2z-1)}\right) = \frac{i}{2}\lim_{z\to 0}\frac{(z-2)(2z-1)4z^3-(z^4+1)(4z-5)}{(z-2)^2(2z-1)^2} = \frac{5i}{8}$

$\displaystyle\text{Res}\left(\tfrac{1}{2}\right) = \lim_{z\to\frac{1}{2}}\frac{i(z^4+1)(z-\frac{1}{2})}{2z^2(z-2)(2z-1)} = -\frac{17i}{24}$

$\displaystyle\int_0^{2\pi}\frac{\cos 2\theta}{5-4\cos\theta}d\theta = 2\pi i\left[\text{Res}(0)+\text{Res}\left(\tfrac{1}{2}\right)\right] = 2\pi i\left(\frac{5i}{8}-\frac{17i}{24}\right) = \frac{\pi}{6}$

(e)



Set

$$I = \int_{-\infty}^{\infty} \frac{dx}{(x^2 + a^2)^2},$$

and integrate

$$f(z) = \frac{1}{(z^2 + a^2)^2} = \frac{1}{(z - ia)^2 (z + ia)^2}$$

along the contour in Figure VII.2.2.
Residue at a double pole at $z_1 = ia$, where by Rule 2,

$$\text{Res}\left[\frac{1}{(z^2 + a^2)^2}, ia\right] = \lim_{z \to ia} \frac{d}{dz} \frac{1}{(z + ia)^2} = \left.\frac{-2}{(z + ia)^3}\right|_{z=ia} = -\frac{i}{4a^3}.$$

Integrate along γ_1, and let $R \to \infty$. This gives

$$\int_{\gamma_1} f(z)\,dz = \int_{-R}^{R} \frac{dx}{(x^2+a^2)^2} \to \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2} = I.$$

Integrate along γ_2, and let $R \to \infty$. This gives

$$\left| \int_{\gamma_2} f(z)\,dz \right| \leq \frac{1}{(R^2-a^2)^2} \cdot \pi R \sim \frac{\pi}{R^3} \to 0.$$

Using the *Residue Theorem* and letting $R \to \infty$, we obtain that that

$$I + 0 = 2\pi i \cdot \left(-\frac{i}{4a^3} \right),$$

and therefore

$$\int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^2} = \frac{\pi}{2a^3}.$$